

BOARD PROPOSAL · JULY 2026

ThaibaHive

The Unified Digital Workplace

A comprehensive staff platform for **Thaiba Garden Group of Institutions**

connecting 800–1,000 employees across all campuses into one secure, intelligent, and easy-to-use system.

Proposed by: Media Department

Building on: MediaHive (production-proven foundation)

For: Board of Directors — Thaiba Garden Group of Institutions

Executive Summary

ThaibaHive is not another HR application. It is the **digital operating system** for Thaiba Garden Group of Institutions—a unified workplace that replaces spreadsheets, paper registers, and fragmented WhatsApp communication with a single platform for attendance, tasks, leaves, reports, approvals, communication, resources, and analytics.

Built on the proven, production-tested foundation of **MediaHive** (already in daily use at the Media Department), ThaibaHive eliminates development risk and delivers a battle-tested platform at a fraction of the cost of commercial ERP or HRMS alternatives.

The approach is **pilot-first and phased**: a representative group of 30–50 staff across roles will validate each feature set in real conditions. After pilot success is confirmed against strict metrics, the full platform launches to all staff across all institutions simultaneously—maximizing adoption momentum while minimizing operational risk.

₹2.14L

First-year investment

₹58

Per staff / year ongoing

4

Months to full launch

30+

Admin hours saved /
week

Why ThaibaHive Matters

Institutional Outcomes, Not Just Features

✓ Save Thousands of Administrative Hours

Automated attendance, digital leave workflows, and standardized reporting eliminate manual paperwork across all campuses.

✓ Improve Staff Accountability

Real-time attendance, task tracking, and report compliance make responsibilities visible and measurable.

✓ Faster Decision Making

Live dashboards give leadership instant visibility into attendance, task completion, leave trends, and department health.

✓ Reduce Paper & Redundancy

Digital forms, circulars, approvals, and reports eliminate printing, filing, and duplicate data entry.

✓ One Platform for Every Institution

All campuses, departments, and sub-departments share one unified system with role-based access.

✓ Digital Transformation Initiative

ThaibaHive positions the institution group as a technology-forward organization, setting the standard for educational administrative excellence.

The Problem

Currently, the institution manages staff across **multiple campuses, departments, schools, and NGOs** with fragmented, manual processes:

Current State

- No centralized attendance system
- Manual work reporting (or none at all)
- Fragmented communication across WhatsApp groups, notice boards, word of mouth
- Paper-based leave with no tracking
- Verbal task assignment — no accountability
- No visibility into staff availability
- Scattered approval processes
- No centralized asset tracking

With ThaibaHive

- Automated NFC / QR attendance, real-time dashboards
- Standardized daily + task-based reports
- Central announcements, notices, department boards, emergency broadcast
- Digital apply → approve workflow with balance tracking
- Assigned, tracked, time-stamped tasks with Kanban boards
- Live presence & availability status
- Unified approval center (leave, purchase, expense, IT, overtime)
- QR-tagged assets with full service history

The Solution: ThaibaHive Features

A web + mobile platform with **28+ integrated features** organized into tiers by priority and value.

Tier 1 — Essential (Core Platform)

Core

Staff Attendance

NFC tag / QR code check-in & check-out.
Configurable shift hours per department, institution, or individual. Late detection and real-time dashboard.

Core

Task Management

Assign tasks to individuals or departments. Kanban board with To Do, In Progress, Review, Completed. Priorities, due dates, comments, file attachments.

Core

Leave Management

Digital apply → approve workflow. Leave type configuration, balance tracking, department-wise overview, auto-update on approval.

Core

Daily Work Reports

End-of-day summaries and task-based reporting. Configurable per staff role. Submission compliance tracking.

Core

Staff Directory

Searchable directory with department, role, institution, contact information. Complete staff profile with portfolio.

Core

Staff Profile Portfolio

Full HR profile: qualifications, certificates, experience, skills, languages, emergency contacts, Aadhaar/PAN, bank details, joining documents, contract expiry, teaching subjects.

Core

Staff Timeline

Complete history per staff member — attendance, leaves, tasks, reports, promotions, department changes, warnings, appreciations — all in one chronological view.

Core

Approval Center

Unified inbox for all approvals — leaves, purchases, expenses, IT requests, document approvals, overtime, duty changes, event approvals.

Tier 2 — High Value

High

Announcements & Notices

Broadcast to all staff, specific departments, or institutions. Read receipts and priority levels.

High

Events Calendar

Institution-wide and department events with RSVP, calendar views, and reminders.

High

Circulars & Document Repository

Central storage for policies, circulars, forms, and official documents. Searchable by category, date, department.

High

IT Help Desk

Raise, track, and resolve IT support tickets with comment threads, priority levels, and assignment tracking.

High

Asset Management

QR-tagged institutional assets (cameras, laptops, projectors, furniture, ACs). Purchase date, warranty, service history, assigned person. NFC scan for quick check-in/out.

High

Digital Forms Builder

Admins create custom forms without coding — hostel inspection, vehicle request, trip approval, purchase request, campus inspection. Integrated approval workflows.

High

Accounts & Financial Tracking

Income/expense tracking per institution. Categories (tuition, salary, maintenance, etc). Transaction ledger with filters. Summary dashboards.

High

Executive Dashboard

Board-level view: staff present, leave statistics, task completion, department productivity, institution comparison, open tickets, maintenance requests.

Tier 3 — Productivity & Engagement

High

Expense Claims

Staff submit bills and travel expenses with photo upload. Approval workflow included. Track reimbursement status.

High

Purchase Requests

Request items through workflow: Staff → HOD → Accounts → Purchase → Approval. Full audit trail.

High

Resource Booking

Book auditoriums, cameras, drones, projectors, DSLRs, laptops, sports grounds, conference halls. Avoid double booking with live calendar.

High

Vehicle Management

Fleet registry, driver assignment, fuel logs, maintenance schedules, insurance expiry alerts, trip requests, fuel efficiency (km/L) calculation.

High

Visitor Management

Reception registers visitors, scans ID, notifies host staff, prints visitor pass, records exit. Security log with full audit trail.

High

Staff Polls & Surveys

Quick feedback collection on proposals, event attendance, training needs. Live bar chart results.

High

Recognition & Badges

Employee of the Month, Excellent Teacher, Team Player, Innovation Award. Peer recognition tokens visible on profiles. Birthday & work anniversary auto-reminders.

High

Canteen / Meal Management

Staff notify skip or guests per meal (breakfast/lunch/dinner). Headcount summary per meal type. Reduces food waste across campuses.

Tier 4 — Advanced & Future

Future

Internal Chat

Direct messages, department groups, file sharing, voice notes, read receipts. Reduces dependency on personal WhatsApp groups.

Future

Shift Planner & Substitution

Auto-suggest substitute staff when leave is approved. Shift swapping for support staff with HOD approval. Recurring shift templates.

Future

Staff Availability Status

Like Microsoft Teams: Available, Busy, In Meeting, On Leave, Working Off Campus, Travelling. Visible across the platform.

Future

Maintenance Tickets

Report facility issues (fan, leakage, electrical, internet). Assign technicians, track completion, SLA monitoring.

Future

Performance Reviews

Quarterly reviews with goals, ratings, manager comments, and growth plans. Links to task completion and report compliance data.

Future

Training Portal

Track training sessions, attendance, certificates, and required courses. Onboarding modules with mandatory quizzes.

Future

Department Health Score

Each department scored on attendance, task completion, report submission, response time, leave trends. Identifies areas needing support.

Future

Workflow Automation

Event-driven automation: "When leave is approved → notify HOD → update attendance → inform security → update calendar." Zero manual orchestration.

AI & Smart Features

High

AI Assistant

Natural language queries: "Who is on leave tomorrow?", "Show pending tasks for Media Department.", "Find all Maths teachers." No menus required.

High

AI Daily Summary

Every morning: who is absent, important meetings, pending approvals, overdue tasks, birthdays, critical alerts. Delivered automatically.

Future

AI Report Generator

Generate monthly department reports, attendance summaries, and management meeting reports in one click.

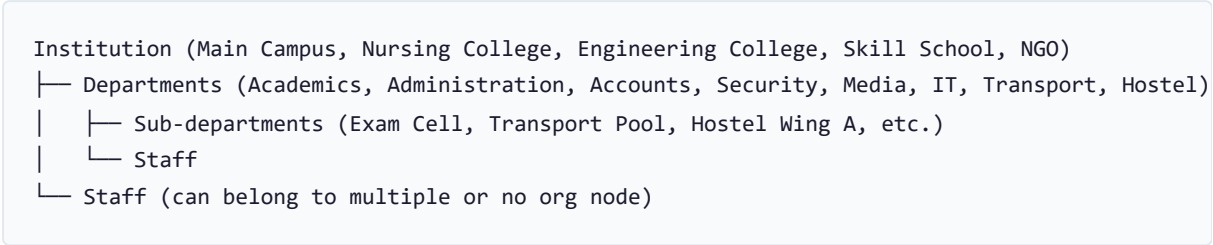
Future

Reminder Engine

Automatically reminds about expiring IDs, probation reviews, contract renewals, pending reports, unapproved leave, incomplete tasks.

Organizational Structure

ThaibaHive supports flexible org hierarchies reflecting the institution group's multi-campus, multi-department reality:



Key flexibility: Staff can belong to multiple departments/institutions, have no department affiliation (central admin/support), and each org node can have independent shift/attendance policies.

Technology Stack

Layer	Technology	Why
Web Frontend	Next.js + React + Tailwind CSS	Proven from MediaHive production
Mobile App	Flutter (Android + iOS)	Offline-first, cross-platform
Backend & Database	Supabase + PostgreSQL	Scalable, real-time, secure
Authentication	Supabase Auth + Google OAuth	Enterprise-grade security
Attendance Hardware	NFC tags + QR codes	Low-cost, reliable, reusable
Offline Sync	IndexedDB (web) / Hive (mobile)	Works without internet
Hosting	Scalable cloud	Auto-scales with user base

Security & Data Protection

Security is built in from day one:	
Concern	Implementation

Access Control	Row-Level Security (RLS) in PostgreSQL — every query is scoped to the user's role and permissions
Role-Based Permissions	4-tier RBAC: Super Admin, Admin, HOD, Staff — granular per-module permissions
Data Encryption	Encrypted in transit (TLS) and at rest (AES-256)
Authentication	Password hashing (bcryptjs) + JWT sessions with httpOnly cookies
Audit Logs	Every action logged with timestamp, user, and details
Daily Backups	Automated database backups with 30-day retention
Disaster Recovery	Point-in-time recovery, multi-region option
Compliance	Designed to meet local data protection regulations

Return on Investment

The question is not whether we can afford ThaibaHive. It is whether we can afford to continue without it.

Quantified Savings

Area	Current Cost	With ThaibaHive	Annual Savings
Attendance processing	~15 hrs/week (manual registers)	Automated, real-time	~₹1.2L in admin time
Leave management	~8 hrs/week (paper forms, tracking)	Digital workflow, auto-balance	~₹0.6L in admin time
Task coordination	~10 hrs/week (chasing, meetings)	Kanban, auto-assign, tracking	~₹0.8L in productive time
Reporting	~6 hrs/week (compiling, formatting)	Auto-generated from live data	~₹0.5L in admin time
Communication	Fragmented, untracked	Central, trackable, searchable	~₹0.3L in wasted effort
Printing & stationery	~₹2,500/month	Digital forms & approvals	~₹0.3L direct savings
Help desk (manual)	~4 hrs/week (phone, walk-ins)	Self-service ticketing	~₹0.3L in support time
Estimated Total Annual Savings			~₹4.0L

Based on blended staff cost of ₹250/hr. First-year investment of ₹2.14L is recovered in approximately 6 months.

Qualitative Benefits

- **Leave processing: Days → Hours** — Digital workflow eliminates paper routing
- **Accountability: None → Measurable** — Every task, report, and attendance is tracked
- **Decision making: Gut feel → Data-driven** — Live dashboards for leadership
- **Staff satisfaction: Higher** — Self-service, transparency, recognition

- **Institutional reputation: Enhanced** — Digital-first, technology-forward

Budget

Item	Cost (₹)	Notes
One-Time Setup Costs		
NFC tags (10 entry points × 2 campuses)	12,500	Reusable, durable tags
QR code printed materials	2,500	Weatherproof signage
Contingency buffer (15%)	2,250	Broken tags, unforeseen hardware
Subtotal (One-Time)	17,250	
Annual Recurring Costs		
Cloud hosting (Supabase + Next.js)	72,000	Scalable for 1,000 users
Mobile app distribution	0	Free on Play Store / App Store
Domain & SSL	2,500	Annual renewal
NFC tag replacements	2,500	~10% annual loss/damage
Subtotal (Annual)	77,000	
Development & Implementation		
Development (reusing MediaHive)	0	In-house by Media Department
Training & change management	20,000	Pilot training, materials, support
Pilot deployment & validation	15,000	100 hrs × 3 staff
Subtotal (Implementation)	35,000	
Year 1 Total	₹1,29,250	
Annual Recurring (Year 2+)	₹77,000	~₹58/staff/year

Key advantages: No annual licence fees. Institution owns the platform. No vendor lock-in. Fully customizable. Future modules reuse the same infrastructure at minimal additional cost.

Risk & Mitigation

Risk	Likelihood	Impact	Mitigation
Staff resistance to adoption	Medium	High	Pilot group builds advocates. Training sessions. Incentives for early adopters. Clear communication of benefits.
Internet outages affecting attendance	Medium	Medium	Offline mode: NFC/QR data cached locally, syncs when connected. Paper backup for critical check-ins.
Data migration errors	Low	Medium	Phased migration with validation checks. Parallel run during pilot. Rollback plan documented.
Software bugs in production	Medium	Medium	4-phase build with testing at every phase. Pilot validation gates. Hotfix deployment within 24 hours.
Security breach / data leak	Low	Critical	RLS, encryption, audit logs, auto-backups, access reviews. No sensitive data stored unnecessarily.
Support staff without smartphones	Medium	Medium	Dedicated tablets at entry points. QR-based check-in at common areas. NFC tags at turnstiles.
Long-term maintenance gap	Medium	High	IT Department trained during build. Documentation & runbooks created. Transition plan from Media to IT.
Scope creep delaying launch	Medium	Medium	Phased rollout with clear feature freeze per phase. Future features captured in roadmap, not current build.

Rollout Strategy: 5 Phases

PHASE 1 — FOUNDATION

Weeks 1–6

Attendance, Directory & Access Control

- NFC/QR attendance with configurable shifts
- Staff directory with org hierarchy & portfolio profiles
- Role-based access control (Super Admin, Admin, HOD, Staff)
- Auth system with secure login

PHASE 2 — PRODUCTIVITY

Weeks 7–10

Tasks, Reports, Leaves & Approvals

- Task management with Kanban board
- Daily work reports & task-based reporting
- Leave management with approval workflow & balance tracking
- Unified approval center

PHASE 3 — COMMUNICATION

Weeks 11–13

Announcements, Events, Circulars & Polls

- Announcements & departmental notice boards
- Events calendar with RSVP
- Circulars & document repository
- Staff polls & surveys

PHASE 4 — SERVICES

Weeks 14–17

Help Desk, Assets, Visitors, Recognition

- IT help desk with ticket tracking
- Resource booking (rooms, equipment)
- Asset management with QR tagging
- Visitor management system
- Staff recognition, badges, birthdays

PHASE 5 — FINANCE & OPERATIONS

Weeks 18–22

Accounts, Vehicles, Canteen, Digital Forms

- Income/expense tracking per institution
- Expense claims & purchase requests with approval workflow
- Vehicle management (fleet, bookings, fuel logs)
- Canteen meal notifications & headcount
- Digital forms builder
- Executive dashboard & department health scores

INSTITUTION-WIDE LAUNCH

Week 23–24

Big Bang Launch — All Features, All Staff, All Campuses

- Rollout to all 800–1,000 staff
- Training sessions for all roles
- Support & monitoring period
- Collect feedback for Phase 6 improvements

Pilot Plan

Pilot Group

30–50 staff from the main campus, representing a balanced cross-section of roles:

- **Management / Administration** (3–5) — Validate dashboards, approvals, reporting
- **Department Heads / HODs** (5–8) — Validate task assignment, leave approval, team oversight
- **Teaching / Academic staff** (10–15) — Validate attendance, reports, leave, tasks
- **Support staff** (8–12) — Security, housekeeping, transport — validate NFC/QR check-in
- **Media / IT Department** (4–5) — Technical validation, feedback, bug reporting

Device Policy

Not all staff own smartphones capable of running the app. ThaibaHive addresses this with a tiered approach:

- **BYOD (Bring Your Own Device):** Most staff use personal smartphones
- **Institution-provided tablets:** Dedicated devices stationed at key entry points for QR/NFC check-in
- **NFC tags at turnstiles:** Quick tap check-in without needing a phone
- **Paper fallback:** Manual check-in option during the transition period

Pilot Validation Criteria

Metric	Target
Attendance check-in compliance	≥ 90%
Daily report submission rate (eligible staff)	≥ 85%
User satisfaction survey	≥ 4 / 5
Critical bugs	0
Features validated by real usage	100%
Task assignment & tracking adoption	≥ 70% of HODs assigning tasks
Leave approval processing time	≤ 4 hours (target: same-day)

Post-Launch Success Metrics

Metric	Target
Attendance compliance (all institutions)	95%+
Report submission rate	90%+
Task completion rate (on time)	80%+
Leave processing time	Reduced from days to hours
Staff satisfaction (after 3 months)	80%+ positive
Monthly active users	90%+ of all staff

Long-Term Vision & Roadmap

ThaibaHive is designed as a **platform for decades**, not a project for months. The same infrastructure supports unlimited future modules:

"ThaibaHive is more than a staff management application—it is the digital operating system for Thaiba Garden Group of Institutions. By connecting people, processes, and information on one platform, it lays the foundation for smarter decisions, greater accountability, and sustainable growth across every campus."

Future Expansion Modules

Visitor Management

Full visitor lifecycle: pre-registration, ID scan, host notification, pass printing, exit logging.

Asset Lifecycle Management

QR-tagged assets with warranty, service history, assignment, geofenced NFC check-in/out.

Vehicle & Transport Tracking

GPS tracking, route optimization, student transport management, fuel efficiency analytics.

Inventory & Procurement

Stock management, reorder alerts, supplier management, purchase order workflow.

Payroll Integration

Attendance data feeds into payroll. Payslip portal, PF statements, tax breakdown sheets.

Student Portal

Attendance view, assignments, communication with parents, fee status.

Parent Portal

Real-time updates on student attendance, performance, events, and communications.

AI Analytics & Reports

Predictive analytics for attrition risk, performance trends, and resource optimization.

Learning Management (LMS)

Course materials, assignments, quizzes, progress tracking for academic programs.

Hostel Management

Room allocation, attendance, visitor logs, maintenance, leave tracking for hostel students.

Growth-ready architecture: Every future module runs on the same infrastructure — no new servers, databases, or platforms needed. The incremental cost of adding a module is primarily development time.

Integration Possibilities

ThaibaHive is designed to integrate with existing and future systems:

System	Integration Type
Google Workspace	Calendar sync, single sign-on, Drive document attachment
Microsoft 365	Calendar sync, Teams status, Outlook contact sync
WhatsApp / Telegram	Automated notifications for approvals, reminders, broadcasts
Email (SMTP)	Transactional emails for approvals, notifications, reports
Biometric Devices	Attendance data import from existing biometric systems
Payroll Software	Attendance and leave data export for payroll processing
Existing ERPs	Staff data sync, single source of truth
SMS Gateway	Emergency broadcasts and critical notifications via SMS

Support & Maintenance

Long-term operational ownership is critical for a 1,000-user platform. Our plan:

Phase	Responsible	Activities
Build & Pilot (Months 1–6)	Media Department	Development, pilot support, bug fixes, training
Transition (Months 6–7)	Media → IT Department	Knowledge transfer, documentation handover, runbook creation
Operations (Month 8+)	IT Department	User support, password resets, database backups, monitoring
Enhancements (Ongoing)	Media Department	New features, version upgrades, integration development

Documentation deliverables: Technical architecture guide, admin playbook, user training manual, backup & restore runbook, troubleshooting guide.

Conclusion & Recommendation

Recommendation: Proceed to Project Kickoff.

ThaibaHive represents an **exceptional return on investment**. For a first-year cost of approximately ₹1.3L (less than ₹130 per staff member), the institution gains:

- A unified digital workplace serving all 800–1,000 employees
- 28+ integrated features covering attendance, productivity, communication, operations, and analytics
- AI-powered automation for daily summaries, natural language queries, and report generation
- A platform that scales to 5,000+ staff and 50+ institutions
- Zero vendor lock-in, no annual licence fees, full institutional ownership
- Estimated annual operational savings of ₹4.0L+ (recovering the investment in ~6 months)

The proposal is built on the proven, production-tested MediaHive foundation, eliminating development risk. The pilot-first strategy ensures real-world validation before full rollout. The phased

approach keeps delivery manageable and focused.

ThaibaHive is more than a staff management application—it is the digital operating system for Thaiba Garden Group of Institutions. By connecting people, processes, and information on one platform, it lays the foundation for smarter decisions, greater accountability, and sustainable growth across every campus.

Next Steps

1. **Board approval** to proceed with Project Kickoff
2. **Finalize pilot group** composition across roles and departments
3. **Set up NFC infrastructure** at main campus entry points
4. **Configure org structure**, roles, shift policies, and access permissions
5. **Begin Phase 1 development** using the MediaHive foundation
6. **Deploy pilot** and begin validation within 2 weeks of approval
7. **Schedule board update** at the end of each phase (every 4–6 weeks)

Prepared by the Media Department — Building on the success of MediaHive for the benefit of the entire institution.

Thaiba Garden Group of Institutions — July 2026